# PC Software Control

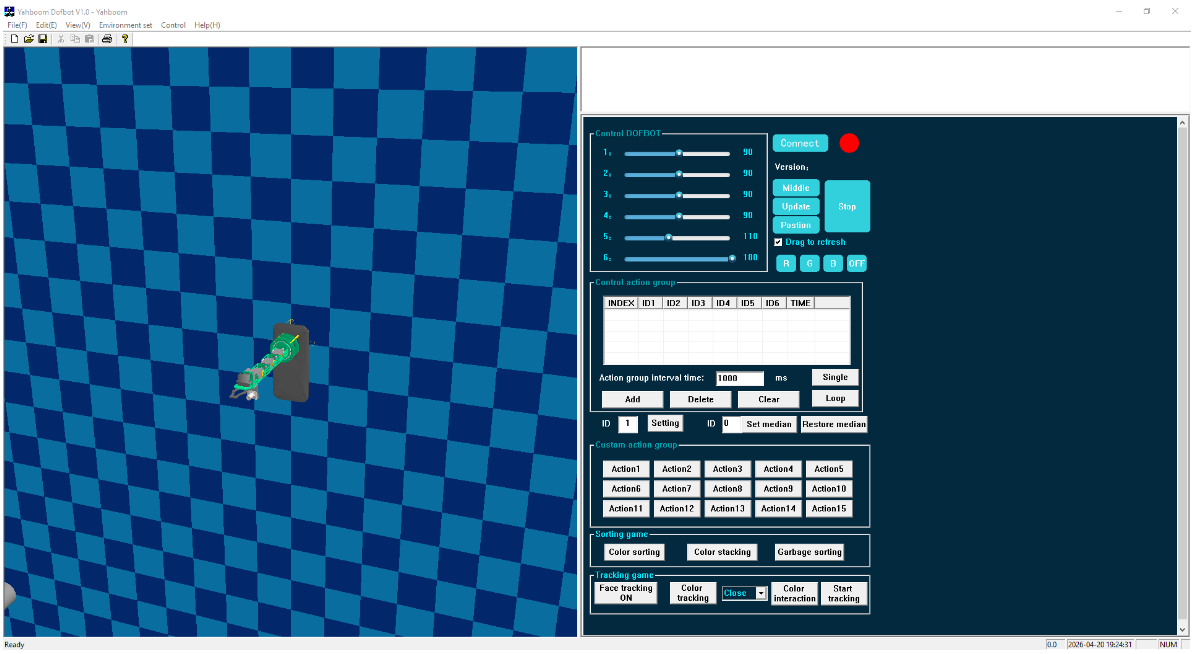
## 1.Start up DOFBOT and connect PC software

1. **Install the PC Software**. Run the installer and follow the prompts to install the Yahboom Dofbot SE robot arm PC software.
2. **Set up peripheral monitoring hardware**. You need to know the robot arm's network IP address, which can be found by logging into the virtual machine and running `ifconfig`.
3. **Start the main program**. Run the main program on the virtual machine. The robot arm will beep three times to indicate it has started normally and is waiting for a connection. Execute the following command:

   python3 ~/rosmaster/YahboomArm.pyc
4. **Launch the PC software**. Open the PC software, ensure the robot arm and computer are on the same local area network, and configure the IP address and port number to connect to the device.
5. **Troubleshooting**
   - **Hardware connection failed**: Hardware is not powered on. Please check if the network is connected.
   - **Software connection failed**: Incorrect IP address or port number. Check if the network is on the same LAN. If the camera feed is not displaying, verify that the USB cable is plugged into the robot arm's USB port.
   - **System error**: Restart the PC software.

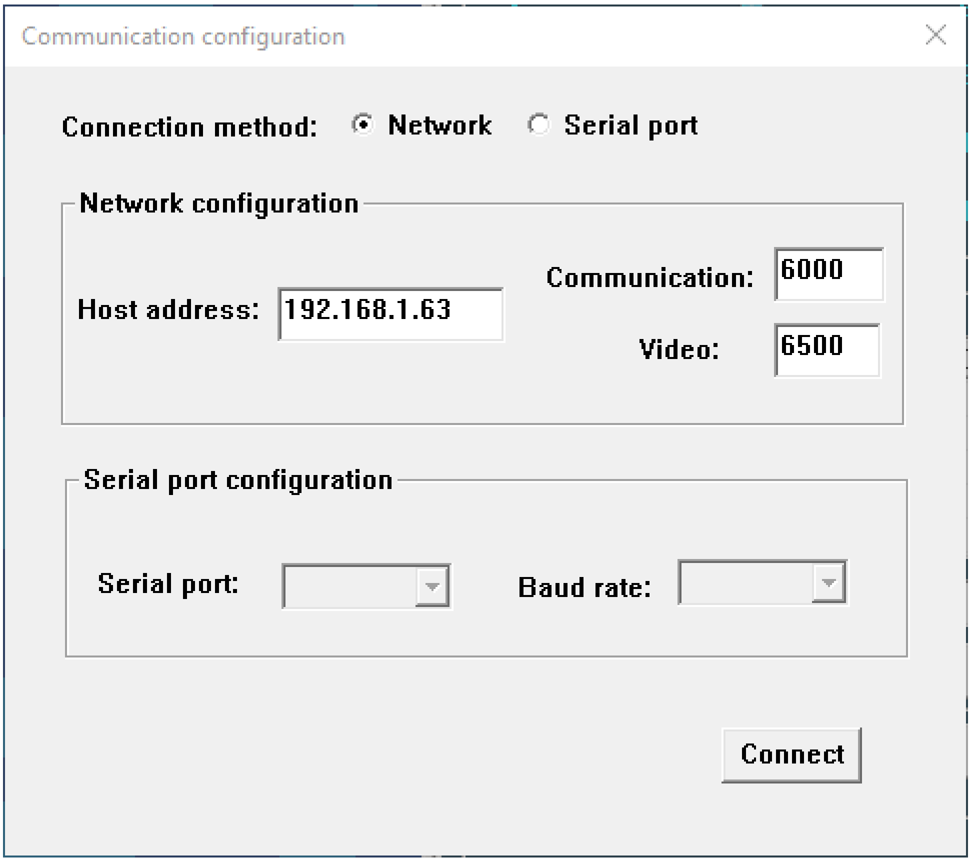
## 2.User Operation Examples

- Open the software. The interface is shown below.

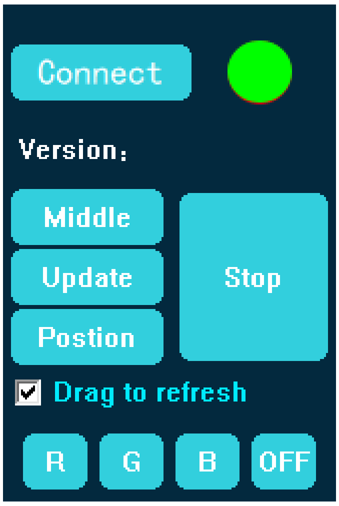



- **Configure connection**

After the hardware starts successfully, click **[Connect]** on the right. The following dialog will appear.



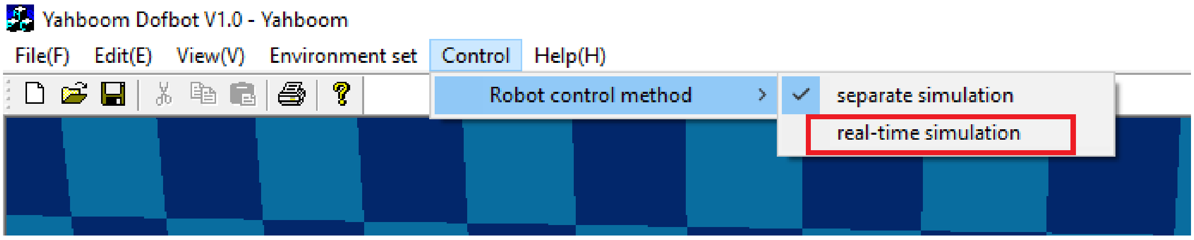

Select the network connection method, enter the robot arm's IP address, communication port, and video port, then click Connect. If the connection is successful, the status indicator on the interface will turn green, as shown below. The video feed will also load.

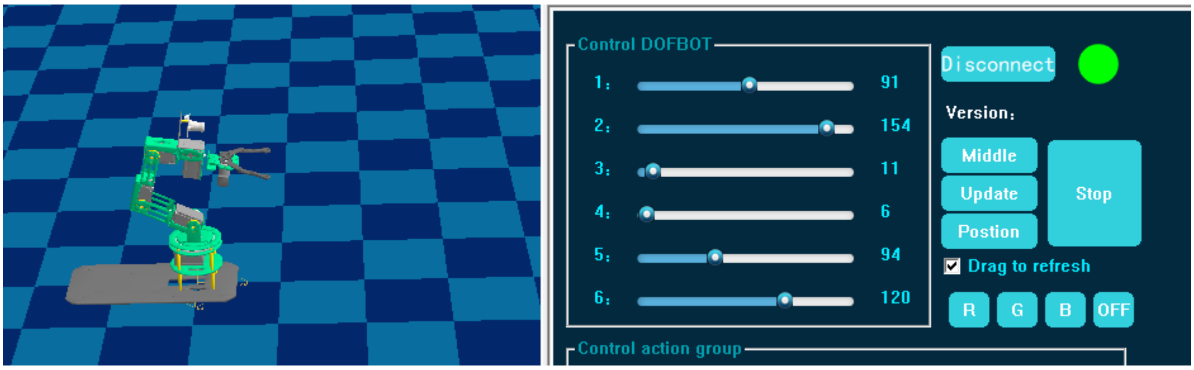



- **Feature demonstrations**

**A: Control DOFBOT**

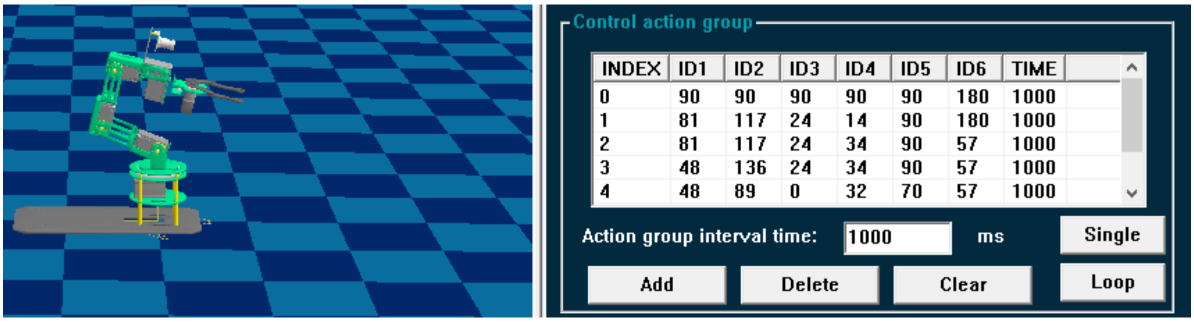
Sliding the slider controls the real-time pose of the 3D simulation on the left interface. By selecting individual simulation or real-time simulation, you can switch whether to send commands to the robot arm during control. You can also use the keyboard to control the multi-DOF robot arm: WASD, TFGH, IJKL control multiple degrees of freedom. Note: When controlling for the first time, please click **[Get Angle]** first. If the angle cannot be obtained, click it again. Wait until the pose on the left 3D interface matches the physical arm before controlling; otherwise, extreme movements may damage the robot arm.



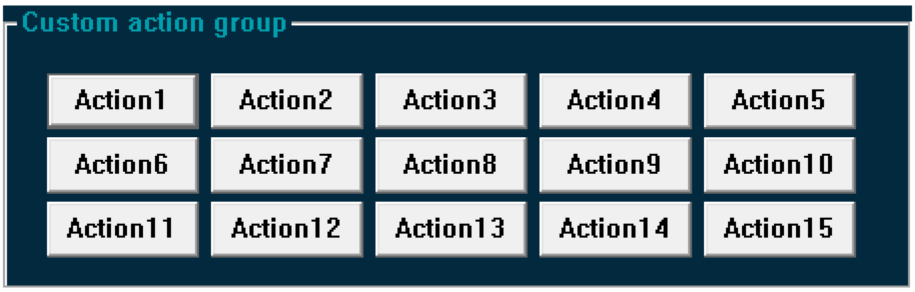

## B: Control action group

In action group editing, drag the sliders and click **[Add Action Group]** to save to the computer. After adding, you can click **[Single]** for single execution or **[Loop]** for continuous execution to run the robot arm. Five groups of actions added are shown below.




## C: Custom action group

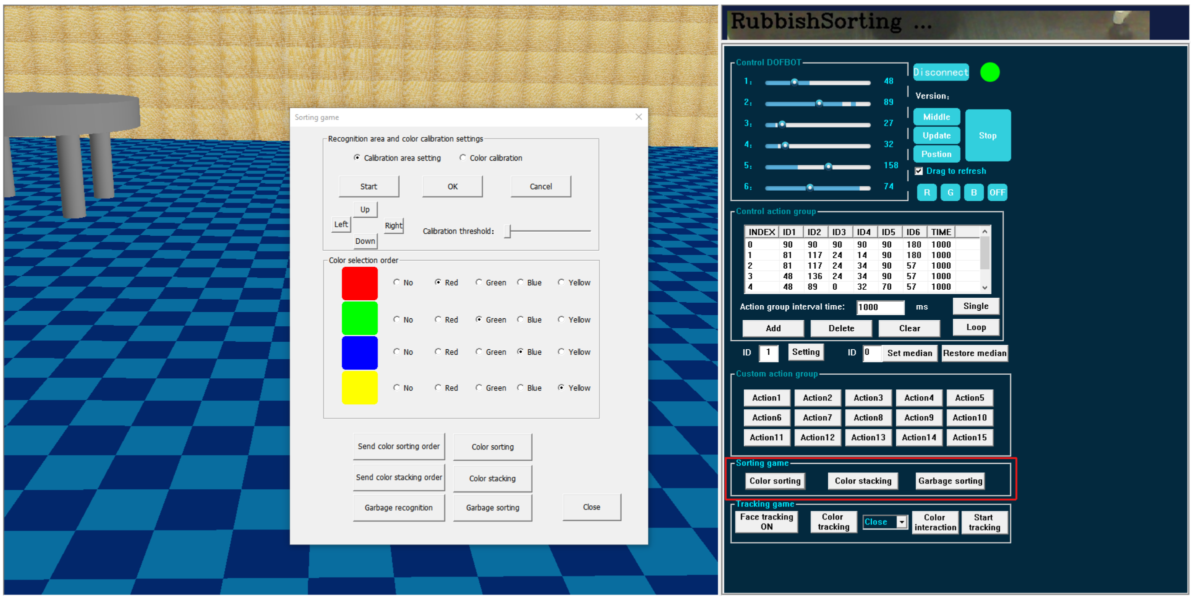
Choose Action X, the robot arm will automatically complete this action, and the robot arm will automatically stop after the action is completed.




## D: Sorting game

Calibration: Click any function in the sorting game box below. A dialog on the left will appear. Install the calibration map correctly. First, you need to set up the calibration area. Click **[Calibration Area Setting]** to enter the calibration area. Click **[Up/Down/Left/Right]** to move so that the entire recognition frame is visible in the camera view. Slide the **[Calibration Threshold]** so that the blue frame encircles the recognition frame in the map. Click **[OK]** to complete calibration.

Gameplay: Select color blocks, then click **[Send Color Sorting Order]** or **[Send Color Stacking Order]**. Wait until the camera recognizes all blocks, then click **[Color Sorting]** or **[Color Stacking]**. For garbage sorting, first click **[Garbage Recognition]**, wait for recognition to complete, then click **[Garbage Sorting]**. The robot arm will place the garbage into the corresponding dustbin according to the sorting category.




**E: Tracking game**

The tracking game is shown below.

【Face tracking】: Click **[Face Tracking ON]**. Look at the robot arm's camera and move slowly. The robot arm will track and follow your face.

【Color tracking】: Select the tracking color from the dropdown on the right, then click **[Color Tracking]**. Place the corresponding color block in front of the robot arm's camera, then move the block. The robot arm will track the color block up, down, left, and right.

【Color interaction】: Click **[Color Interaction]**. Place the color to be learned in front of the camera and position it within the green recognition frame on the screen. Wait until the value above the green frame stabilizes, then click **[Start Tracking]**. The robot arm will track and follow the learned color block.